

Wednesday, February 26

Parameter Interpretation With Log Link Functions

A GLM with a log link function, like a Poisson regression model, has the form

$$\log E(Y_i) = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_k x_{ik},$$

or

$$E(Y_i) = \exp(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_k x_{ik}),$$

which can also be written as a “multiplicative model” of the form

$$E(Y_i) = e^{\beta_0} e^{\beta_1 x_{i1}} e^{\beta_2 x_{i2}} \cdots e^{\beta_k x_{ik}}.$$

Recall that $e^{a+b} = e^a e^b$. For this reason the parameters $\beta_1, \beta_2, \dots, \beta_k$ or linear functions thereof are not interpreted the same way as in the *additive* model

$$E(Y_i) = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_k x_{ik},$$

but they are still relatively easy to interpret in terms of *multiplicative* rather than *additive* changes in $E(Y)$.

Rate Ratios (Quantitative Explanatory Variable)

Consider the model

$$\log E(Y) = \beta_0 + \beta_1 x,$$

and let

$$\log E(Y_a) = \beta_0 + \beta_1(x+1) \quad \text{and} \quad \log E(Y_b) = \beta_0 + \beta_1 x$$

for an arbitrary value of x . Then the *difference* in the log of the expected values is

$$\log E(Y_a) - \log E(Y_b) = \underbrace{\beta_0 + \beta_1(x+1)}_{\log E(Y_a)} - \underbrace{(\beta_0 + \beta_1 x)}_{\log E(Y_b)} = \beta_1,$$

meaning that β_1 is the *additive* change in $\log E(Y)$ per unit increase in x .

Now consider the same model written as

$$E(Y) = e^{\beta_0} e^{\beta_1 x},$$

and let

$$E(Y_a) = e^{\beta_0} e^{\beta_1(x+1)} \quad \text{and} \quad E(Y_b) = e^{\beta_0} e^{\beta_1 x}$$

for an arbitrary value of x . Then the *ratio* of the expected values is

$$\frac{E(Y_a)}{E(Y_b)} = \frac{\overbrace{e^{\beta_0} e^{\beta_1(x+1)}}^{E(Y_a)}}{\underbrace{e^{\beta_0} e^{\beta_1 x}}_{E(Y_b)}} = \frac{e^{\beta_0} e^{\beta_1 x} e^{\beta_1}}{e^{\beta_0} e^{\beta_1 x}} = e^{\beta_1} \Rightarrow E(Y_a) = E(Y_b) e^{\beta_1},$$

so that $E(Y)$ changes by a *factor* of e^{β_1} per unit increase in x . The “exponentiated” parameter, e^{β_1} , is sometimes called a “rate ratio” because it is often the ratio of two rates when the counts are per unit space, time, or something else.

Example: Consider again the *ceriodaphniastrain* data and model.

```
library(trtools)
ceriodaphniastrain$strainf <- factor(ceriodaphniastrain$strain,
  labels = c("a", "b"))
m <- glm(count ~ concentration + strainf,
  family = poisson, data = ceriodaphniastrain) # log link is default
cbind(summary(m)$coefficients, confint(m))
```

	Estimate	Std. Error	z value	Pr(> z)	2.5 %	97.5 %
(Intercept)	4.455	0.0391	113.82	0.00e+00	4.38	4.53
concentration	-1.543	0.0466	-33.11	2.06e-240	-1.63	-1.45
strainfb	-0.275	0.0484	-5.68	1.31e-08	-0.37	-0.18

```
exp(cbind(coef(m), confint(m))) # coef extracts the parameter estimates only
```

		2.5 %	97.5 %
(Intercept)	86.025	79.615	92.817
concentration	0.214	0.195	0.234
strainfb	0.760	0.691	0.835

Note: It only makes sense to apply the exponential function to the point estimates and the endpoints of the confidence interval. A standard error of $e^{\hat{\beta}_1}$ could be obtained, but it is **not** equal to the exponentiated standard error of $\hat{\beta}_1$. A test concerning e^{β_1} can be done using either the confidence interval or by stated the hypotheses in terms of β_1 (e.g., the null hypothesis that $e^{\beta_1} = 1$ is the same as the null hypothesis that $\beta_1 = 0$).

Another approach is to use `lincon` and the `tf` (transformation function) argument.

```
lincon(m, tf = exp)
```

	estimate	lower	upper
(Intercept)	86.025	79.673	92.884
concentration	0.214	0.195	0.234
strainfb	0.760	0.691	0.835

Note that the confidence interval endpoints are not quite the same as what we obtained using `confint`. This is because `confint` and `lincon` use different approaches to confidence intervals (more on that later).

Example: Consider a model for the expected number of matings of African elephants as a function of age.

```
library(Sleuth3)
head(case2201)
```

	Age	Matings
1	27	0
2	28	1
3	28	1
4	28	1
5	28	3
6	29	0

```
m <- glm(Matings ~ Age, family = poisson, data = case2201)
cbind(summary(m)$coefficients, confint(m))
```

	Estimate	Std. Error	z value	Pr(> z)	2.5 %	97.5 %
(Intercept)	-1.5820	0.5446	-2.9	3.68e-03	-2.6667	-0.5289
Age	0.0687	0.0137	5.0	5.81e-07	0.0417	0.0956

```
exp(cbind(m$coefficients, confint(m)))
```

		2.5 %	97.5 %
(Intercept)	0.206	0.0695	0.589
Age	1.071	1.0426	1.100

Percent Change (Quantitative Explanatory Variable)

The *percent change* in the expected response is

$$100\% \times \left[\frac{E(Y_a) - E(Y_b)}{E(Y_b)} \right] = 100\% \times [E(Y_a)/E(Y_b) - 1],$$

where $E(Y_a)$ and $E(Y_b)$ are the expected responses at two different points (a and b) defined in terms of the explanatory variable(s).

1. Note that if this is *positive* then it is a percent *increase*, whereas if it is negative then it is a percent *decrease*.
2. The ratio $E(Y_a)/E(Y_b)$ is the *rate ratio*.

Example: Suppose we have the model $\log E(Y) = \beta_0 + \beta_1 x$ where x is a quantitative variable and $\beta_1 = 0.22$. Then $e^{\beta_1} \approx 1.25$. So when x increases by one unit (i.e., to $x + 1$), — i.e., from $E(Y_b) = e^{\beta_0} e^{\beta_1 x}$ to $E(Y_a) = e^{\beta_0} e^{\beta_1(x+1)}$ then the expected response *increases* by a *factor* of

$$E(Y_a)/E(Y_b) = e^{\beta_1} \approx 1.25,$$

and because

$$100\% \times [1.25 - 1] = 25\%.$$

we can say that it *increases* by 25%.

Example: Consider again the model for the elephant mating data.

```
m <- glm(Matings ~ Age, family = poisson, data = case2201)
exp(cbind(m$coefficients, confint(m)))
```

		2.5 %	97.5 %
(Intercept)	0.206	0.0695	0.589
Age	1.071	1.0426	1.100

The percent change in the expected count per unit (year) increase in Age is approximately $100\%(1.07 - 1) = 7\%$ (i.e., a 7% increase).

Example: Suppose we have the model $\log E(Y) = \beta_0 + \beta_1 x$ where x is a quantitative variable and $\beta_1 = -0.22$. Then $e^{\beta_1} \approx 0.8$. So when x increases by one unit (i.e., to $x + 1$), — i.e., from $E(Y_b) = e^{\beta_0} e^{\beta_1 x}$ to $E(Y_a) = e^{\beta_0} e^{\beta_1(x+1)}$ then the expected response *decreases* by a *factor* of

$$E(Y_a)/E(Y_b) = e^{\beta_1} \approx 0.8,$$

or because

$$100\% \times [0.8 - 1] = -20\%$$

we can say that it *decreases* by 20%.

Example: Consider again the model for the ceriodaphniastrain data.

```
m <- glm(count ~ concentration + strainf, family = poisson, data = ceriodaphniastrain)
exp(cbind(coef(m), confint(m)))
```

		2.5 %	97.5 %
(Intercept)	86.025	79.615	92.817
concentration	0.214	0.195	0.234
strainfb	0.760	0.691	0.835

The percent change in the expected count per unit increase in concentration is approximately $100\%(0.21 - 1) = -79\%$ (i.e., a 79% decrease or reduction).

Rate Ratios (Categorical Explanatory Variable)

Consider the model

$$\log E(Y) = \beta_0 + \beta_1 x, \text{ or, equivalently, } E(Y) = e^{\beta_0} e^{\beta_1 x},$$

where

$$x = \begin{cases} 1, & \text{if the observation is in group } a, \\ 0, & \text{if the observation is in group } b. \end{cases}$$

Then

$$E(Y) = \begin{cases} e^{\beta_0} e^{\beta_1}, & \text{if the observation is in group } a, \\ e^{\beta_0}, & \text{if the observation is in group } b. \end{cases}$$

Let

$$E(Y_a) = e^{\beta_0} e^{\beta_1} \quad \text{and} \quad E(Y_b) = e^{\beta_0}.$$

Then the *ratio* of the expected values is

$$\frac{E(Y_a)}{E(Y_b)} = \frac{e^{\beta_0} e^{\beta_1}}{e^{\beta_0}} = e^{\beta_1} \Leftrightarrow E(Y_a) = E(Y_b) e^{\beta_1}$$

so that $E(Y_a)$ is e^{β_1} times that of $E(Y_b)$. Also

$$\frac{E(Y_b)}{E(Y_a)} = \frac{e^{\beta_0}}{e^{\beta_0} e^{\beta_1}} = \frac{1}{e^{\beta_1}} = e^{-\beta_1}.$$

so that $E(Y_b)$ is $1/e^{\beta_1}$ times that of $E(Y_a)$.

Example: Consider again the `ceriodaphniastrain` data and model.

```
m <- glm(count ~ concentration + strainfb,
  family = poisson, data = ceriodaphniastrain)
cbind(summary(m)$coefficients, confint(m))
```

	Estimate	Std. Error	z value	Pr(> z)	2.5 %	97.5 %
(Intercept)	4.455	0.0391	113.82	0.00e+00	4.38	4.53
concentration	-1.543	0.0466	-33.11	2.06e-240	-1.63	-1.45
strainfb	-0.275	0.0484	-5.68	1.31e-08	-0.37	-0.18

```
exp(cbind(coef(m), confint(m)))
```

		2.5 %	97.5 %
(Intercept)	86.025	79.615	92.817
concentration	0.214	0.195	0.234
strainfb	0.760	0.691	0.835

Alternatively we can parameterize the model.

```
ceriodaphniastrain$strainfb <- relevel(ceriodaphniastrain$strainfb, ref = "b")
m <- glm(count ~ concentration + strainfb,
  family = poisson, data = ceriodaphniastrain)
cbind(summary(m)$coefficients, confint(m))
```

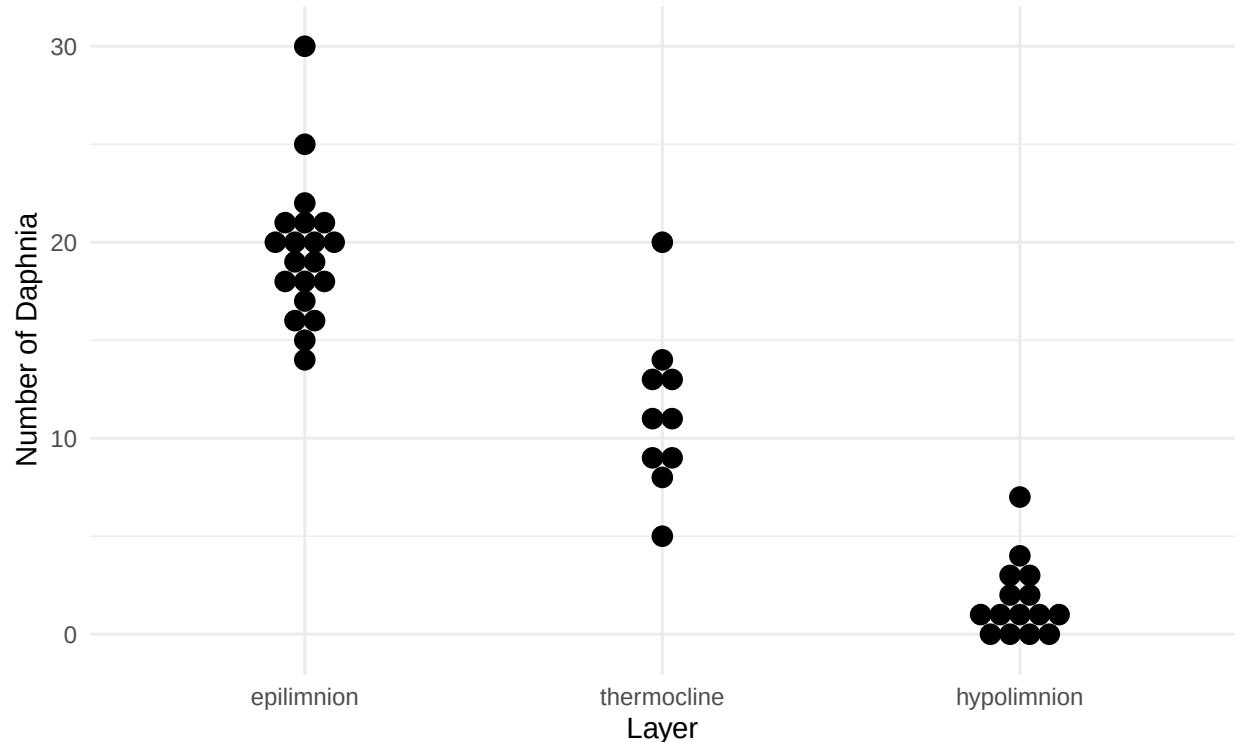
	Estimate	Std. Error	z value	Pr(> z)	2.5 %	97.5 %
(Intercept)	4.180	0.0430	97.14	0.00e+00	4.09	4.26
concentration	-1.543	0.0466	-33.11	2.06e-240	-1.63	-1.45
strainfa	0.275	0.0484	5.68	1.31e-08	0.18	0.37

```
exp(cbind(coef(m), confint(m)))
```

		2.5 %	97.5 %
(Intercept)	65.344	60.008	71.034
concentration	0.214	0.195	0.234
strainfa	1.316	1.198	1.448

Example: Consider these data from a stratified random sampling design and a Poisson regression model.

```
library(trtools)
library(ggplot2)
p <- ggplot(daphniastrat, aes(x = layer, y = count)) +
  geom_dotplot(binaxis = "y", binwidth = 1, stackdir = "center") +
  labs(x = "Layer", y = "Number of Daphnia") + theme_minimal()
plot(p)
```



```
daphniastrat$layer <- relevel(daphniastrat$layer, ref = "thermocline")
m <- glm(count ~ layer, family = poisson, data = daphniastrat)
summary(m)$coefficients
```

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	2.425	0.0941	25.78	1.65e-146
layerepilimnion	0.546	0.1068	5.11	3.27e-07
layerhypolimnion	-1.875	0.2175	-8.62	6.74e-18

```
exp(cbind(coef(m), confint(m)))
```

		2.5 %	97.5 %
(Intercept)	11.36	10.25	12.57
layerepilimnion	1.73	1.51	1.97
layerhypolimnion	0.16	0.14	0.18

(Intercept)	11.300	9.3425	13.513
layerepilimnion	1.726	1.4050	2.137
layerhypolimnion	0.153	0.0981	0.231

Percent Larger/Smaller (Categorical Explanatory Variable)

The *percent change* in the expected response is

$$100\% \times \left[\frac{E(Y_a) - E(Y_b)}{E(Y_b)} \right] = 100\% \times [E(Y_a)/E(Y_b) - 1],$$

where $E(Y_a)$ and $E(Y_b)$ are the expected responses at two different points (a and b) defined in terms of the explanatory variable(s).

1. Note that if this is *positive* then $E(Y_a)$ is that percent *larger* than $E(Y_b)$, whereas if this is *negative* then $E(Y_b)$ is that percent *smaller* than $E(Y_a)$.
2. The ratio $E(Y_a)/E(Y_b)$ is the *rate ratio*.

Example: Suppose we have the model $\log E(Y) = \beta_0 + \beta_1 x$ where x is an indicator variable for category a and $\beta_1 = 0.22$. Then $e^{\beta_1} \approx 1.25$, $E(Y_a) = e^{\beta_0} e^{\beta_1}$ and $E(Y_b) = e^{\beta_0}$, and $E(Y_a)$ is about 1.25 times *larger* than $E(Y_b)$ because

$$E(Y_a)/E(Y_b) = e^{\beta_1} \approx 1.25,$$

and because

$$100\% \times [1.25 - 1] = 25\%.$$

we can say that $E(Y_a)$ is about 25% *larger* than $E(Y_b)$.

Example: Suppose we have the model $\log E(Y) = \beta_0 + \beta_1 x$ where x is an indicator variable for category a and $\beta_1 = -0.22$. Then $e^{\beta_1} \approx 0.8$, $E(Y_a) = e^{\beta_0} e^{\beta_1}$ and $E(Y_b) = e^{\beta_0}$, and $E(Y_a)$ is about 0.8 times *smaller* than $E(Y_b)$ because

$$E(Y_a)/E(Y_b) = e^{\beta_1} \approx 0.8,$$

and because

$$100\% \times [0.8 - 1] = -20\%.$$

we can say that $E(Y_a)$ is about 20% *smaller* than $E(Y_b)$.

Example: Consider again the model for the daphnia data.

```
exp(cbind(coef(m), confint(m)))
```

		2.5 %	97.5 %
(Intercept)	11.300	9.3425	13.513
layerepilimnion	1.726	1.4050	2.137
layerhypolimnion	0.153	0.0981	0.231

The expected number of daphnia per liter in the epilimnion layer is estimated to be about $100\%(1.73-1) = 73\%$ more than in the thermocline layer. And because $100\%(0.15-1) = -85\%$ we estimate that the the expected number of daphia per liter in the hypolimnion layer is 85% less than it is in the thermocline layer.

Contrasts With Log Link Functions

With a log link function a “contrast” as produced by the `contrast` function has the general form

$$\log E(Y_a) - \log E(Y_b) = \log \left[\frac{E(Y_a)}{E(Y_b)} \right],$$

where the indices a and b denote specific values of the explanatory variables. If we apply the exponential function to the contrast then it becomes

$$\exp[\log E(Y_a) - \log E(Y_b)] = \frac{E(Y_a)}{E(Y_b)},$$

So applying the exponential function to contrasts allows us to interpret them as ratios.

Example: Consider again the stratified random sampling design. Suppose we want to compare the epilimnion and thermocline layers to the hypolimnion layer. We can use `contrast` and apply the exponential function (`exp` in R) through the argument `tf` (for “transformation function”). Note that this function is only applied to the estimates and the confidence intervals.

```
trtools::contrast(m,
  a = list(layer = c("epilimnion", "thermocline")),
  b = list(layer = "hypolimnion"),
  cnames = c("epil vs hypo", "therm vs hypo"))
```

	estimate	se	lower	upper	tvalue	df	pvalue
epil vs hypo	2.42	0.203	2.02	2.82	11.95	Inf	6.52e-33
therm vs hypo	1.87	0.218	1.45	2.30	8.62	Inf	6.74e-18

```
trtools::contrast(m,
  a = list(layer = c("epilimnion", "thermocline")),
  b = list(layer = "hypolimnion"),
  cnames = c("epil/hypo", "therm/hypo"), tf = exp)
```

	estimate	lower	upper
epil/hypo	11.25	7.56	16.73
therm/hypo	6.52	4.26	9.98

The following gives us inferences for the *logarithm* of the expected count for each layer.

```
trtools::contrast(m, a = list(layer = c("epilimnion", "thermocline", "hypolimnion")),
  cnames = c("epilimnion", "thermocline", "hypolimnion"))
```

	estimate	se	lower	upper	tvalue	df	pvalue
epilimnion	2.97	0.0506	2.871	3.070	58.7	Inf	0.00e+00
thermocline	2.42	0.0941	2.240	2.609	25.8	Inf	1.65e-146
hypolimnion	0.55	0.1961	0.166	0.934	2.8	Inf	5.04e-03

To produce the estimates of the expected counts we need to apply the exponential function.

```
trtools::contrast(m, a = list(layer = c("epilimnion", "thermocline", "hypolimnion")),
  cnames = c("epilimnion", "thermocline", "hypolimnion"), tf = exp)
```

	estimate	lower	upper
epilimnion	19.50	17.66	21.53
thermocline	11.30	9.40	13.59
hypolimnion	1.73	1.18	2.55

The `emmeans` package can also produce inferences for expected counts and rate ratios for categorical explanatory variables if we specify `type = "response"`.

```
library(emmeans)
emmeans(m, ~layer, type = "response")
```

layer	rate	SE	df	asyp.LCL	asyp.UCL
thermocline	11.30	1.060	Inf	9.40	13.59
epilimnion	19.50	0.987	Inf	17.66	21.53

```
hypolimnion 1.73 0.340 Inf 1.18 2.55
```

Confidence level used: 0.95

Intervals are back-transformed from the log scale

```
pairs(emmeans(m, ~layer), type = "response", adjust = "none", infer = TRUE)
```

contrast	ratio	SE	df	asympt.LCL	asympt.UCL	null	z.ratio	p.value
thermocline / epilimnion	0.58	0.062	Inf	0.47	0.71	1	-5.110	<.0001
thermocline / hypolimnion	6.52	1.420	Inf	4.26	9.98	1	8.620	<.0001
epilimnion / hypolimnion	11.25	2.280	Inf	7.56	16.73	1	11.950	<.0001

Confidence level used: 0.95

Intervals are back-transformed from the log scale

Tests are performed on the log scale

Suppose we want the rate ratios comparing the epilimnion and thermocline with the hypolimnion layer.

```
contrast(emmeans(m, ~layer, type = "response"),
  method = "trt.vs.ctrl", ref = 3, type = "response", infer = TRUE)
```

contrast	ratio	SE	df	asympt.LCL	asympt.UCL	null	z.ratio	p.value
thermocline / hypolimnion	6.52	1.42	Inf	4.02	10.6	1	8.620	<.0001
epilimnion / hypolimnion	11.25	2.28	Inf	7.17	17.6	1	11.950	<.0001

Confidence level used: 0.95

Conf-level adjustment: dunnett method for 2 estimates

Intervals are back-transformed from the log scale

P value adjustment: dunnett method for 2 tests

Tests are performed on the log scale

Another tool that you can use if you want inferences about the expected response is the `glmint` function from the `trtools` package.

```
d <- data.frame(layer = c("epilimnion", "thermocline", "hypolimnion"))
glmint(m, newdata = d) # syntax similar to predict and nlsint
```

	fit	low	upp
1	19.50	17.66	21.53
2	11.30	9.40	13.59
3	1.73	1.18	2.55

Example: Consider again the model for the `ceriodaphniastrain` data. Consider first the effect of increasing concentration by one percent.

```
m <- glm(count ~ concentration + strainf,
  family = poisson, data = ceriodaphniastrain)
summary(m)$coefficients
```

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	4.180	0.0430	97.14	0.00e+00
concentration	-1.543	0.0466	-33.11	2.06e-240
strainfa	0.275	0.0484	5.68	1.31e-08

```
exp(cbind(coef(m), confint(m)))
```

		2.5 %	97.5 %
(Intercept)	65.344	60.008	71.034
concentration	0.214	0.195	0.234


```
strainfa      1.316  1.198  1.448
```

We can estimate the rate ratio for a one unit increase in concentration for each strain.

```
trtools::contrast(m,
  a = list(concentration = 1, strainf = c("a","b")),
  b = list(concentration = 0, strainf = c("a","b")),
  cnames = c("a","b"), tf = exp)
```

```
      estimate lower upper
a      0.214 0.195 0.234
b      0.214 0.195 0.234
```

Here is how we can do that with the **emmeans** package. This statement will give us the expected response for concentrations one unit apart for each strain.

```
emmeans(m, ~concentration|strainf,
  at = list(concentration = c(1,0)), type = "response")
```

```
strainf = b:
  concentration rate    SE  df asymp.LCL asymp.UCL
      1 14.0 0.61 Inf      12.8      15.2
      0 65.3 2.81 Inf      60.1      71.1
```

```
strainf = a:
  concentration rate    SE  df asymp.LCL asymp.UCL
      1 18.4 0.73 Inf      17.0      19.9
      0 86.0 3.37 Inf      79.7      92.9
```

Confidence level used: 0.95

Intervals are back-transformed from the log scale

Now we can compare them.

```
pairs(emmeans(m, ~concentration|strainf, at = list(concentration = c(1,0)),
  type = "response"), infer = TRUE)
```

```
strainf = b:
  contrast                ratio    SE  df asymp.LCL asymp.UCL null z.ratio p.value
concentration1 / concentration0 0.214 0.00996 Inf      0.195      0.234    1 -33.100 <.0001
```

```
strainf = a:
  contrast                ratio    SE  df asymp.LCL asymp.UCL null z.ratio p.value
concentration1 / concentration0 0.214 0.00996 Inf      0.195      0.234    1 -33.100 <.0001
```

Confidence level used: 0.95

Intervals are back-transformed from the log scale

Tests are performed on the log scale

We can estimate the rate ratio comparing the strains at difference concentrations.

```
trtools::contrast(m,
  a = list(concentration = c(0, 1, 2), strainf = "a"),
  b = list(concentration = c(0, 1, 2), strainf = "b"),
  cnames = c("0%", "1%", "2%"), tf = exp)
```

```
      estimate lower upper
0%      1.32    1.2  1.45
1%      1.32    1.2  1.45
```

```
2%      1.32    1.2  1.45
```

We can also use `contrast` to estimate the expected count for, say, strain `a` at different concentration values.

```
trtools::contrast(m, a = list(concentration = c(0, 1, 2), strainf = "a"),
  cnames = c("0%", "1%", "2%"), tf = exp)
```

```
      estimate lower upper
0%      86.03 79.67 92.88
1%      18.39 17.01 19.87
2%       3.93  3.38  4.57
```

We can also use the `emmeans` package for inferences about expected counts and rate ratios for categorical explanatory variables.

```
library(emmeans)
emmeans(m, ~ strainf, type = "response",
  at = list(concentration = 0))
```

```
strainf rate    SE  df asymp.LCL asymp.UCL
b         65.3 2.81 Inf      60.1      71.1
a         86.0 3.37 Inf      79.7      92.9
```

Confidence level used: 0.95

Intervals are back-transformed from the log scale

```
pairs(emmeans(m, ~ strainf, type = "response",
  at = list(concentration = 0)), reverse = TRUE)
```

```
contrast ratio      SE  df null z.ratio p.value
a / b       1.32 0.0637 Inf    1  5.680 <.0001
```

Tests are performed on the log scale

Now suppose we add an interaction between concentration and strain.

```
m <- glm(count ~ concentration + strainf + concentration:strainf,
  family = poisson, data = ceriodaphniastrain)
summary(m)$coefficients
```

```
              Estimate Std. Error z value Pr(>|z|)
(Intercept)      4.144    0.0510   81.25 0.00e+00
concentration     -1.473    0.0701  -21.01 4.80e-98
strainfa          0.337    0.0670    5.02 5.11e-07
concentration:strainfa -0.125    0.0939   -1.34 1.82e-01
```

```
trtools::contrast(m,
  a = list(concentration = 1, strainf = c("a", "b")),
  b = list(concentration = 0, strainf = c("a", "b")),
  cnames = c("a", "b"), tf = exp)
```

```
      estimate lower upper
a       0.202 0.179 0.229
b       0.229 0.200 0.263
```

```
trtools::contrast(m,
  a = list(concentration = c(0, 1, 2), strainf = "a"),
  b = list(concentration = c(0, 1, 2), strainf = "b"),
  cnames = c("0%", "1%", "2%"), tf = exp)
```

	estimate	lower	upper
0%	1.40	1.228	1.60
1%	1.24	1.082	1.41
2%	1.09	0.813	1.46

Now the rate ratio for concentration depends on strain and the rate ratio for strain depends on concentration when there is an interaction term.